

MCD411A B. Tech. Project

Progress Report 1

Submission Due: 21-Aug-2026, 06:00 PM

Title of the project: *Neural-Network Based KKT-Solver*

Tick (✓) Area of the project:

Design	Industrial ✓	Production	Thermofluids
--------	--------------	------------	--------------

Student Name with Entry number:

Student Name	<i>SAHITYA RANKAWAT</i>	<i>MANAV GUPTA</i>	<i>JEET ANAND</i>
Entry number	<i>2023ME21131</i>	<i>2023ME20733</i>	<i>2023ME20874</i>

Supervisor(s) Name: *KARTIKEY SHARMA*

Abstract (100 words only): Briefly describe the project, expected outcomes

The aim of this project is to develop an optimization solver by means of neural-networks. The approach involves constructing a pipeline in which 'Pyomo' receives the optm problem together with their constraints, and converts them the equations to stdo file, by which a loss function is then formulated on the basis of Karush-Kuhn-Tucker (KKT) conditions. The NN will train using this KKT loss to produce the optimal primal/dual variables, without the need of any labeled data. The anticipated result will be a fully functional solver pipeline which is to be tested against MIPLIB problems for benchmarking and improving the accuracy and the architecture

Objectives: (2-4 bullet points)

- Set up a pipeline using Pyomo to create algebraic KKT equations of the optm. problem
- Develop a neural network architecture that acts as a solver by minimizing a custom loss fⁿ.
- Benchmark the network's accuracy and performance against standard solvers using MIPLIB problems.

Progress/ Activities (till now): (3-5 bullet points)

- Completed literature review on KKT-informed neural networks.
- Started learning & setting up Pyomo to extract algebraic equations for the pipeline.
- Understanding the procedure to create custom loss function according to the input.

Declaration:

I/We have met our supervisor on 19/08/26 (Date) at 2-3 P.M (Time) and discussed the report before submission.

<i>Sahitya</i>	<i>Jeet/Manav</i>	<i>Manav</i>
(Student 1 Signature)	(Student 2 Signature)	(Student 3 Signature)

Performance Assessment: {To be filled in by the Supervisor}

Unsatisfactory	Incremental	Satisfactory
----------------	-------------	--------------

K. S.
(Supervisor Signature)